

FaultBridge

A code-switched Nigerian telco support agent that grounds answers in verified operational data, resolves eligible faults, and turns repeated unresolved complaints into unconfirmed incidents for NOC review.

EVALUATION QUESTION

Can a voice agent understand Hausa-English, Igbo-English, Nigerian Pidgin-English, and Yoruba-English while choosing the correct tools, protecting caller data, and avoiding invented network facts?

400 natural clips

4 ASR systems

68.63 minutes

351 source groups

55.0%

Sahara equal-language WER, the strongest aggregate on the frozen natural-speech panel.

Natural code-switched ASR

We selected 100 hash-frozen AfriSwitch clips per language, balanced across code-mixing, switch count, duration, and source groups. Every model received identical 16 kHz PCM16 audio. Failed and empty outputs stayed in the denominator. Confidence intervals use 2,000 deterministic source-clustered bootstrap samples.

Language pair	Sahara	SBPN	OmniASR	Faster-Whisper
Hausa-English	38.9%	45.1%	68.0%	98.3%
Igbo-English	62.1%	65.7%	79.9%	85.0%
Pidgin-English	35.6%	39.3%	55.7%	51.5%
Yoruba-English	83.6%	82.1%	81.4%	91.2%
Equal-language WER	55.0%	58.1%	71.2%	81.5%

Paired finding. Sahara's WER advantage over SBPN is statistically supported for Hausa, Igbo, and Pidgin. The Yoruba interval crosses zero, so its apparent model ordering is inconclusive.

Honest failure accounting. Empty-output rates were Sahara 10.5%, SBPN 8.8%, OmniASR 4.5%, and Faster-Whisper 4.0%. Eight difficult clips were empty for every model.

SPEECH QUALITY MUST SURVIVE THE AGENT

TTS fidelity, task completion, and privacy

200/200

Sahara TTS outputs generated successfully

648

real model + PostgreSQL agent executions

0%

critical safety failure rate

SAHARA CODE-SWITCHED TTS

One hundred genuine code-switched AfriSwitch texts, 25 per language, were synthesized once with female and male Sahara voices. All outputs were valid 22.05 kHz mono PCM16 WAVs, totaling 1,672.29 seconds with no clipped samples. Three independent ASR families judged every output.

Independent judge	WER	Extra words	Transcript loss	Segment loss
SBPN Multilingual Base	39.7%	2.2%	7.6%	27.7%
Meta OmniASR CTC 300M v2	51.3%	2.1%	9.4%	36.8%
Faster-Whisper large-v3-turbo	58.6%	3.3%	12.7%	39.6%

These are automatic audit signals, not confirmed TTS defects. Judge disagreement is large, so a frozen 40-file bilingual listening audit is prepared before any MOS or human-confirmed hallucination claim.

EXECUTABLE AGENT EVALUATION

The 48-scenario capability panel covered 12 resolution and safety paths per language. Together-hosted Llama 3.3 70B, deterministic policy, real typed tools, and isolated PostgreSQL state passed **288/288** gold and controlled-stress executions. Gold analysis latency was 1.20 s p50 / 2.25 s p95.

24-utterance named-ASR panel	Strict success	Assertion accuracy	Capability retained
Gold transcript	100.0%	100.0%	100.0%
Sahara ASR	100.0%	100.0%	100.0%
SBPN	95.8%	99.2%	95.8%
OmniASR	87.5%	98.4%	87.5%
Faster-Whisper	79.2%	97.6%	79.2%

FAILURE ANALYSIS

All 27 strict failures came from nine stable scenario/provider cells where ASR lost a symptom or language cue. No run leaked declared PII or promoted an unconfirmed incident as fact.

PRIVACY GATE

100 synthetic cases, 100 independently labelled phone, email, account/SIM, and numeric spans: 100% exact typed-span precision, recall, and F1; zero post-redaction leakages within scope.

Metrics, reproducibility, and limits

METRIC DEFINITIONS

- **WER** = substitutions + deletions + insertions, divided by reference words. Lower is better and values may exceed 100%.
- **Hallucination / transcript loss** use insertion and deletion rates as automatic TTS indicators.
- **Segment loss** is the share of tagged language spans with no correctly aligned word.
- **Strict task success** requires every analysis, tool-trace, response, state, and privacy assertion to pass.
- **pass@1 / pass@3 / pass³** mean first-run, any-of-three, and all-three success.
- **Capability retention** divides ASR-transcript task success by gold-transcript success.

REPRODUCIBLE EVIDENCE

- Frozen manifests bind every sample and generated WAV to SHA-256.
- Raw JSONL is append-only and retains failures, provider IDs, parameters, model versions, code commit, lock hash, and latency.
- ``make benchmark-verify-asr-evidence``, ``-tts-evidence``, and ``-agent-evidence`` regenerate public aggregates byte for byte.
- Private ASR evidence: Tiamz/faultbridge-asr-benchmark-evidence, revision ef320d7f...fe00.

MEASURED ROUTING DECISION

Use Sahara for all four language pairs in the submission. No alternative currently combines a statistically supported paired WER advantage, $\leq 1\%$ provider-failure evidence, comparable deployed p95 latency, and zero critical downstream failures. The evidence-gated router is implemented, but multi-ASR routing stays off until those conditions pass.

RESPONSIBLE DEPLOYMENT BOUNDARY

Consent and privacy. Declined consent bypasses speech and model providers. Accepted transcripts are redacted before the LLM, database, tools, logs, or UI; caller IDs are HMAC-pseudonymized.

Grounding. Verified incidents and account records enter through authenticated internal APIs. Missing values stay unknown.

Agent authority. The LLM extracts a constrained complaint schema. Deterministic policy authorizes typed tools, and every tool input/output is audited.

Crowd discovery. Three distinct callers may create an *unconfirmed* candidate incident; only an operator can verify it.

LIMITATIONS

The natural ASR panel is difficult research speech, not a population prevalence sample. The named-ASR panel uses one Sahara female synthetic voice and may favor Sahara recognition. Automatic TTS judges cannot replace bilingual listening. The privacy claim excludes names, street addresses, and fully word-spelled numbers. API, CPU, and GPU timing environments are not directly comparable. FaultBridge still requires authoritative NOC/CRM integrations, a telco security review, matched live-stream latency, and human call trials before production.

Sources and artifacts. AfriSwitch dataset card: huggingface.co/datasets/intronhealth/AfriSwitch · Intron speech API: docs.voice.intron.io · RAI speech evaluation: isca-archive.org/interspeech_2025/rai25_interspeech.pdf · Full methods and exact tables: ``docs/ASR_BENCHMARK_REPORT.md``, ``docs/TTS_BENCHMARK_REPORT.md``, ``docs/AGENT_BENCHMARK_REPORT.md``, and ``docs/PRIVACY_BENCHMARK_REPORT.md``.